

MINI PROJECT

SMART POULTRY MONITORING SYSTEM

Members

1. Elvis King'ori
2. Delvan Mucheru
3. Beverly Langat

OBJECTIVE

The main goal of this project was to build a cheap, automatic climate control system for chicken houses using an ESP32-C3 and MicroPython.

Instead of just checking the weather and ringing an alarm, this system actually fixes the problem. It watches the temperature and humidity using a DHT22 sensor and automatically turns on fans, heaters, or humidifiers to keep the birds comfortable, safe, and productive.

REQUIREMENTS

Hardware Requirements

- **ESP32-C3 Microcontroller Development Board:** This is the brain of the project. It handles all the processing, timing loops, and GPIO control.
- **DHT22 Temperature & Humidity Sensor:** Used for checking real-time poultry room conditions.
- **16x2 LCD Display with I2C Module:** Local screen interface. The I2C module is required because it reduces the wiring from 16 data pins down to just 2 pins (SDA and SCL).

Indicators & Actuator Simulation

We used different colored LEDs to represent our heavy-duty environmental actuators:

- **1 Green LED:** Displays the **Global Safe State** (Everything is running normally).
- **1 Red LED:** Simulates the **Infrared Heat Lamp / Heater**.
- **1 Blue LED:** Simulates the **Humidifier (HMDF) / Water Sprayer**.
- **1 White LED:** Simulates the **Ventilation Fan**.
- **1 Active Buzzer (5V):** Sounds the audible safety warning alarm whenever a crisis state is hit.

Support Infrastructure.

- **4 Resistors (220Ω):** Connected in series with the anodes of the LEDs to limit current and prevent blowing out the ESP32-C3 GPIO pins.
- **Breadboard:** For layout assembly and plugging in components during testing.
- **Jumper Wires (Male-to-Male & Female-to-Male):** For hooking up the sensor, LCD, and LEDs to the board rails.

SYSTEM DESCRIPTION

The project is split into three simple parts: Inputs, Processing, and Outputs.

- **Input:** A **DHT22 sensor** reads the room temperature and humidity every 2 seconds.

- **Processing:** The **ESP32-C3 microcontroller** runs the MicroPython code, processes the sensor data, and decides which equipment needs to turn on.
- **Outputs: 16x2 LCD Screen:** Too much text makes the screen look messy. To fix this, the screen automatically switches what it shows every 3 seconds (Temperature info first, then Humidity info next, then it displays system condition as the 3rd cycle).
 - **Green LED:** The "All Clear" light. It stays ON only when both temperature and humidity are perfectly normal.
 - **Red LED:** Acts as the **Heat Lamp**, it lights when the temperature gets too cold.
 - **Blue LED:** Acts as the **Humidifier (HMDF)**, it lights when the air is too dry.
 - **White LED:** Acts as the **Ventilation Fan**. It pulls out hot air if it's too hot, or clears out damp air if it's too humid.
 - **Buzzer:** Sounds an alarm when both temperature and humidity are not in normal conditions.

FUNCTIONAL LOGIC

The system has set safety rules:

- ✓ **Safe Temperature** is 20.0°C to 29.9°C
- ✓ **Safe Humidity** is 40% to 70%.

Here is exactly how the system reacts to the environment:

- **Everything is Perfect:** The **Green LED is ON**. The screen cycles between showing NORMAL TEMP and NORMAL HUM. All fans and heaters stay off. LCD displays **ALL NORMAL**
- **Too Hot (Temperature >=30):** Green LED shuts off. The **White LED (Fan)** turns on to cool things down, the buzzer alarms, and the screen says TEMP:H FAN:ON. LCD displays **COOLING ACTV**
- **Too Cold (Temperature <=19.9):** Green LED shuts off. The **Red LED (Heat Lamp)** turns on to warm up the room, the buzzer alarms, and the screen says TEMP:L LAMP:ON. LCD displays **HEATING ACTV**
- **Too Humid (Humidity > 70):** Green LED shuts off. The **White LED (Fan)** turns on to blow out the damp air, the buzzer alarms, and the screen says HUM:H FAN:ON. LCD displays **VENTILATING**
- **Too Dry (Humidity < 40):** Green LED shuts off. The **Blue LED (Humidifier)** turns on to add moisture, the buzzer alarms, and the screen says HUM:L HMDF:ON. LCD displays **HUMIDIFYING**

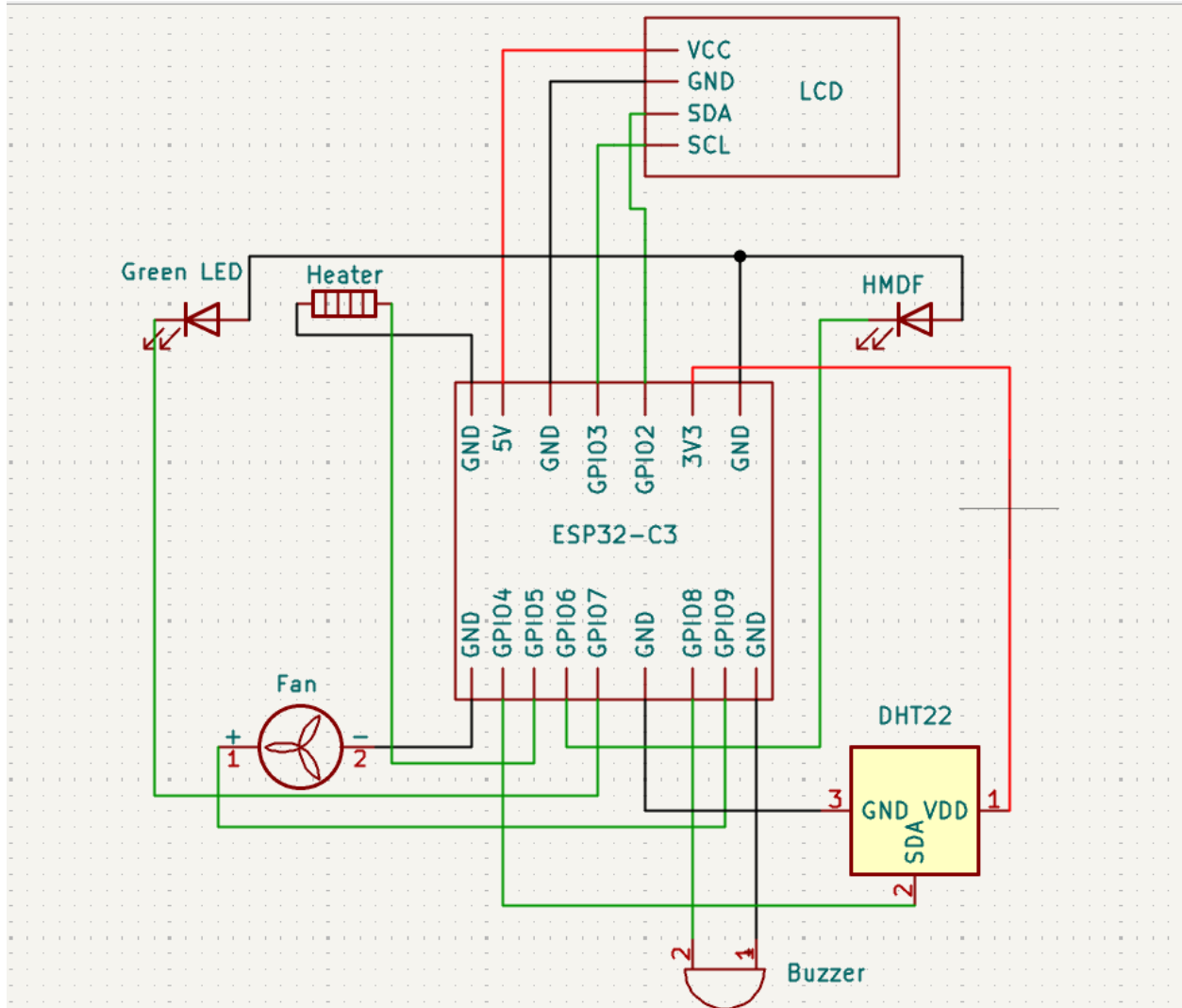
If both temperature and humidity go bad at the same time, the Green LED drops out instantly, the buzzer sounds, and the fan handles both cooling and drying without any code crashes. The LCD then displays **CHECK SYSTEM** warning the farmer that multiple conditions are abnormal and require immediate action.

FUNCTIONALITY

Schematics

The schematics was drawn using Kicad and since most symbols are not inbuilt like the ESP32-C3 and the LCD, we had to create our own symbols from scratch using the symbol editor.

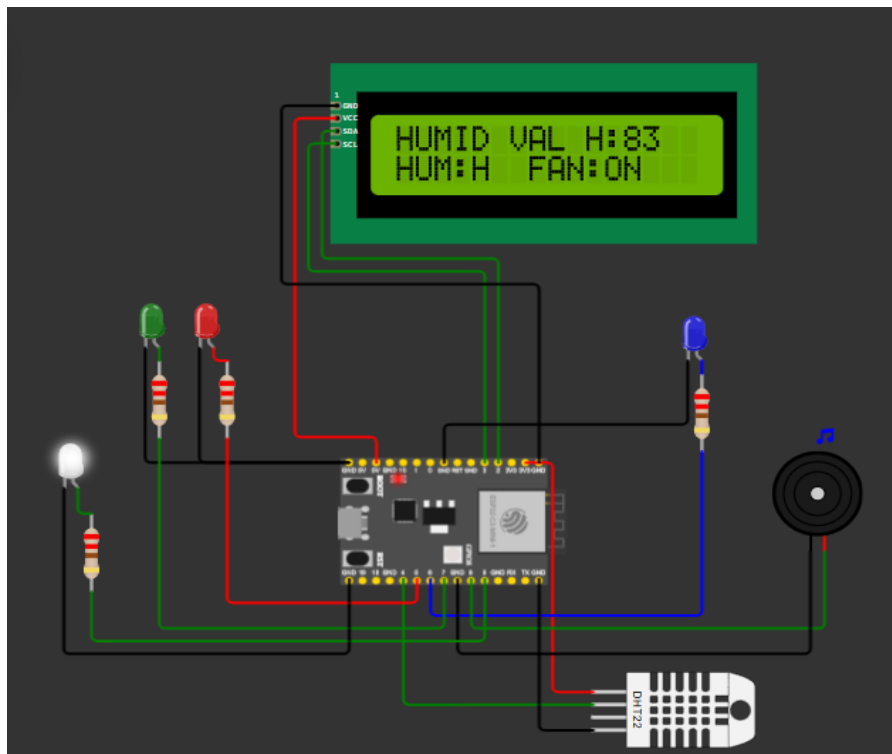
This was our circuit schematic:



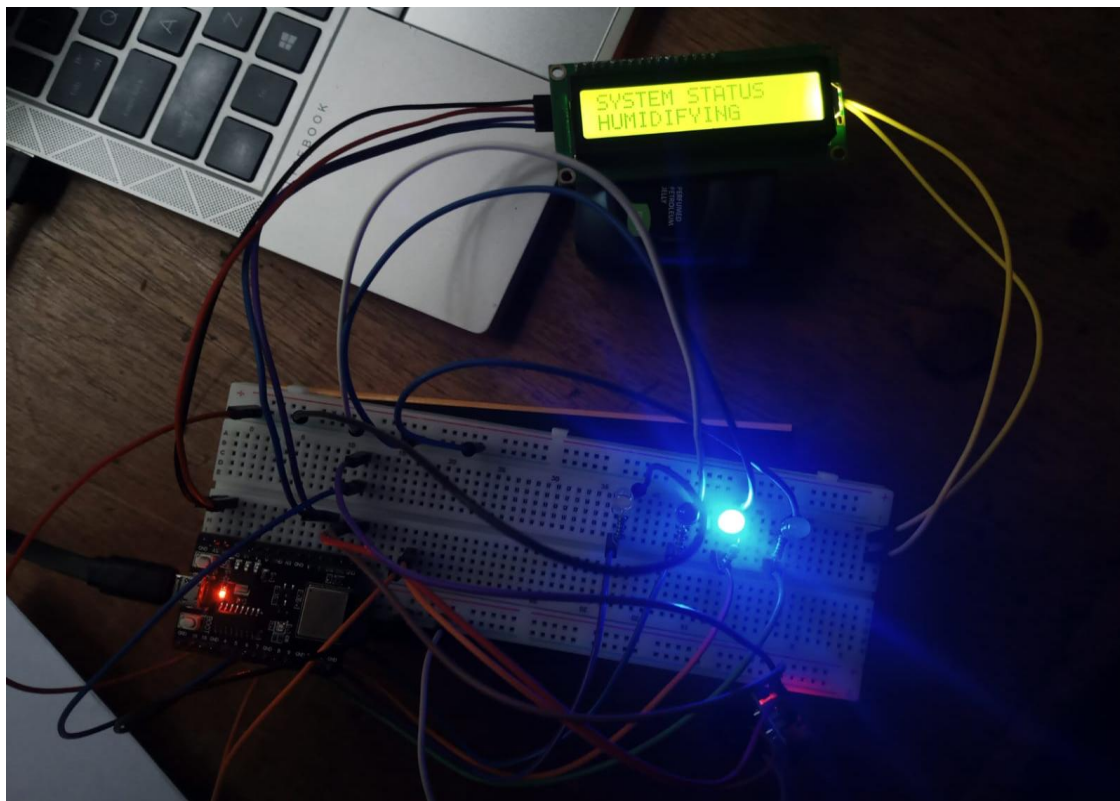
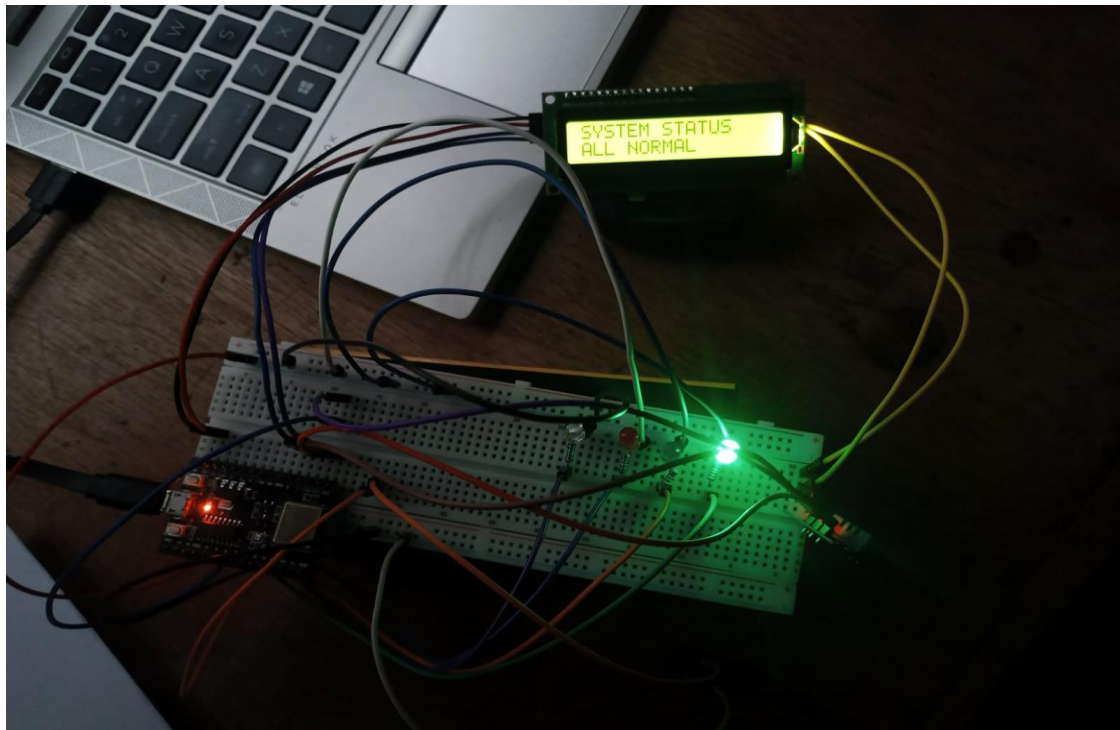
Simulation

We then had to simulate the circuit using Wokwi to get the exact pinouts and the over view of the project structure and to also get a working code.

This is the simulation layout:



Project link: <https://wokwi.com/projects/463375605733684225>



Conclusion

Some limitations of the system are as follows:

- The system was initially built for DHT22 sensor which is much more accurate than the DHT11 we used due to lack of components, DHT11 reacts slightly later to sudden environmental changes.
- The project uses only one temperature/humidity sensor. Conditions may vary across a large poultry house or greenhouse. Other sections may have different conditions that go undetected.
- System stops working during power failure. Since it needs power for proper running, in case of a power outage the system stops working.

Some upgrades we wish to implement when going forward with the project are;

- ✓ Use of real components other than actuators, we simulated fan, humidifier and heat lamp using LEDs, in the future we can implement the project using real components.
- ✓ Use of remote monitoring, currently the LCD displays the conditions of the system which limits remote monitoring, in the future we can implement an IoT web dashboard in which the farmer can monitor the poultry farm.
- ✓ Currently the system only measures temperature and humidity, we can try to add other elements to measure like Ammonia concentration in the poultry farm.
- ✓ We can try to implement a cloud storage for storing data and monitoring trends in the farm.